Deploying Cloud Native Qumulo (CNQ) on AWS with Terraform

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This repository contains Terraform which deploys S3 buckets and a CNQ cluster with 3 or more instances that adhere to the AWS Well-Architected Framework. NOTE: as of version 7.0 this Terraform uses the Qumulo AWS Provider. This greatly simplifies Terraform operations and makes compute elasticity changes even more elegant. To learn more about the provider read the provider docs.

Configure Terraform backend.tf

This Terraform defaults to store state on S3. As such the ./backend.tf must both be configured with the S3 bucket and S3 bucket region you choose to store state in. Note you can't put variables in the backend.tf config, which is why this note is here. If you call this Terraform as a module the backend.tf will be ignored.

Getting Started with Cloud Native Qumulo (CNQ)

For an overview of CNQ, its reference architecture, and limits, see How Cloud Native Qumulo Works and for prerequisites and detailed instructions, see Deploying Cloud Native Qumulo on AWS with Terraform on the Documentation Portal.

Qumulo Core >= 7.9.2.1 is required for this Terraform

> ✅ **Tip:** For help with deployment, configuration, updates, scaling out your cluster, and best practices for high performance, message us on Slack.

Standard Deployment Example

```
module "cloud_native_qumulo" {
  source = "git::https://github.com/Qumulo/aws-terraform-cnq.git?ref=v7.5"
  # ****************************** QUMULO PROVIDER VARIABLES ********************
  #-----------REQUIRED-------------------
  deployment_name = "my-deployment-name"
  ec2_key_pair    = "my-key-pair"
  instance_type   = "i7i.2xlarge"
  region          = "us-west-2"
  subnet_ids      = ["subnet-0123456789abcdef1"]
  vpc_id          = "vpc-0123456789abcdef1"

  #------------OPTIONAL------------------
  additional_security_group_ids = null
  allow_cidrs                   = null
  ami_id                        = null
  cluster_iam_role_arn          = null
  cluster_security_group_id     = null
  kms_key_id                    = null
  permissions_boundary_arn      = null
  provisioner_ami_id            = null
  provisioner_iam_role_arn      = null
  provisioner_instance_type     = null
  provisioner_security_group_id = null
  s3_log_bucket_name            = null
  s3_log_bucket_prefix          = null
  tags = {
    owner        = "owner"
    department   = "department"
    purpose      = "purpose"
    long_running = "true"
  }

  # ***** Qumulo Cluster Variables ******
```

```
  #-----------REQUIRED-------------------
  admin_pwd_or_secrets_arn = "arn:aws:secretsmanager:us-west-2:<aws account
number>:secret:/<secret path>/<secret_name>"
  cluster_name             = "CNQ-HOT"
  cluster_product_type     = "HOT"
  deletion_protection      = true
  node_count               = 3

  #------------OPTIONAL------------------
  audit_logging            = false
  cluster_version          = null
  floating_ip_count        = 12
  ip_v4_or_v6              = "v4"
  nexus_registration_key   = null
  provider_timeout_minutes = 30
  soft_capacity_limit_tb   = null
  storage_class            = null
}

output "outputs_cloud_native_qumulo" {
  value = module.cloud_native_qumulo
}
```

Terraform Documentation

> **Note:** This repository uses documentation generated with Terraform-Docs.

Requirements

Name	Version
terraform	>= 1.11.0
aws	>= 5.32
null	>= 3.1
qumulo	>= 1.4.10
random	>= 3.1

Inputs

Name	Description	Type	Default	Required
additional_security_group_ids	OPTIONAL: Use for adding custom ingress rules beyond those the provider creates	`list(string)`	`null`	no
admin_pwd_or_secrets_arn	Provide either a plaintext administrator password or an AWS Secrets Manager ARN.	`string`	n/a	yes
allow_cidrs	OPTIONAL: CIDR blocks allowed to access the cluster	`list(string)`	`null`	no
ami_id	OPTIONAL: AMI ID for cluster nodes. If omitted, the default Qumulo AMI (Ubuntu 24.04) is used. Supports Ubuntu and RHEL 8, 9, and 10 AMIs. See aws-custom-images.md for details.	`string`	`null`	no

Name	Description	Type	Default	Required
audit_logging	OPTIONAL: Send Qumulo audit logs to AWS CloudWatch logs.	`bool`	`false`	no
cluster_dns_name	OPTIONAL: DNS Name for the cluster. Leave null to automatically pickup the QDNS or NLB DNS name.	`string`	`null`	no
cluster_fqdn	OPTIONAL: Fully qualified domain name for Qumulo DNS	`string`	`null`	no
cluster_iam_role_arn	OPTIONAL: When set, the provider performs no IAM writes for that role — no creation, policy updates, tagging, or deletion — and instead launches instances with your role's instance profile	`string`	`null`	no
cluster_name	Cluster name (2-15 alphanumeric characters). Used as a prefix for AWS resources created for this cluster.	`string`	n/a	yes
cluster_product_type	Cluster storage product type (immutable after creation). HOT: Optimized for frequently accessed data. COLD: Optimized for archival/infrequently accessed data.	`string`	n/a	yes
cluster_security_group_id	OPTIONAL: Bring-your-own security group ID for cluster nodes. When set, the provider will not create or modify a cluster security group; you must pre-create it in the VPC owner account with the rules documented in the Bring-your-own Security Groups guide. Required together with provisioner_security_group_id. Required for RAM-shared subnets where the consumer account cannot create security groups in the owner's VPC.	`string`	`null`	no
cluster_version	OPTIONAL: Qumulo software version. Defaults to latest. Immutable after creation. Upgrade version via cluster UI/API.	`string`	`null`	no
deletion_protection	Enables EC2 Termination protection and prevents	`bool`	`true`	no

Name	Description	Type	Default	Required
	Terraform from destroying non-empty S3 Buckets			
deployment_name	Cluster name (2-15 alphanumeric characters). Used as a prefix for AWS resources created for this cluster.	`string`	n/a	yes
ec2_key_pair	EC2 key pair name for SSH access to cluster nodes	`string`	n/a	yes
floating_ip_count	OPTIONAL: Number of floating IPs. Must be 0, or between 3 and 100. NOT applicable with multi-AZ deployments.	`number`	`3`	no
instance_type	EC2 instance type for the cluster. Prefer i7i, i4i, i7ien. Supported families include m6idn, m6i, m7i, i3en, i4i, i7i, i7ie.	`string`	n/a	yes
ip_v4_or_v6	OPTIONAL: Only change this from default (v4) if connecting to the cluster via IPv6. Not supported for clusters deployed without the Qumulo Terraform Provider.	`string`	`"v4"`	no
kms_key_id	OPTIONAL: KMS key ARN for encrypting AWS services like EBS and S3 (immutable after creation). Qumulo encrypts all data independent of AWS default or KMS keys.	`string`	`null`	no
networking_mode	OPTIONAL: Only change this from default (host_managed) if importing a cluster that was deployed without the Qumulo Terraform Provider. Terraform versions prior to 7.0.	`string`	`"host_managed"`	no
nexus_api_token	OPTIONAL: Qumulo Nexus API token for NeuralProtect	`string`	`null`	no
nexus_registration_key	OPTIONAL: Qumulo Nexus registration key for remote support	`string`	`null`	no
nlb_cross_zone	OPTIONAL: AWS NLB Enable cross-AZ load balancing	`bool`	`false`	no
nlb_dns_client_affinity	OPTIONAL: AWS NLB DNS Client Routing Policy Zonal Affinity	`string`	`"availability_zone_affinity"`	no
nlb_override_subnet_ids	OPTIONAL: Private Subnet ID(s) for NLB if deploying in subnet(s) other than subnet(s) the cluster is deployed in	`list(string)`	`null`	no

Name	Description	Type	Default	Required
nlb_provision	OPTIONAL: Provision an AWS NLB in front of the Qumulo cluster for load balancing and client failover	`bool`	`false`	no
nlb_public	OPTIONAL: Makes the NLB for the cluster public, setting this to true will allow anyone to reach the cluster. Not recommended for production clusters.	`bool`	`false`	no
nlb_stickiness	OPTIONAL: AWS NLB sticky sessions	`bool`	`true`	no
node_count	Number of nodes in the cluster. Valid values: 1 (single node), or 3-24. 1 and 4 not allowed for multi-AZ.	`number`	n/a	yes
node_hooks_files	OPTIONAL: Pre and post userdata hooks files for Cluster Nodes.	`map(string)`	`null`	no
np_deletion_protection	OPTIONAL: Causes Terraform to throw an error upon destroy for the NeuralProtect resource. Safegaurd your NeuralProtect instance. Set to false to destroy.	`bool`	`true`	no
np_instance_type	OPTIONAL: NeuralProtect EC2 instance type.	`string`	`"m6a.xlarge"`	no
np_provision	OPTIOINAL: true/false to enable deployment of NeuralProtect.	`bool`	`false`	no
permissions_boundary_arn	OPTIONAL: IAM permissions boundary ARN applied to cluster and provisioner roles	`string`	`null`	no
provider_timeout_minutes	The total time after which Terraform will abondon the provider deployment of the Qumulo cluster and timeout. In minutes.	`number`	`30`	no
provisioner_ami_id	OPTIONAL: AMI ID for the provisioner instance. Defaults to Ubunti 24.04 AMI.	`string`	`null`	no
provisioner_hooks_files	OPTIONAL: Pre and post userdata hooks files for the provisioner.	`map(string)`	`null`	no
provisioner_iam_role_arn	OPTIONAL: When set, the provider performs no IAM writes for that role — no creation, policy updates, tagging, or deletion — and instead launches instances	`string`	`null`	no

Name	Description	Type	Default	Required
	with your role's instance profile			
provisioner_instance_type	OPTIONAL: EC2 instance type for the provisioner VM (used during deploy operations). Default: m5.xlarge.	`string`	`"m5.xlarge"`	no
provisioner_security_group_id	OPTIONAL: Bring-your-own security group ID for the provisioner instance. Required together with cluster_security_group_id. See cluster_security_group_id.	`string`	`null`	no
r53_second_subnet_id	OPTIONAL: A second subnet ID, in a unique AZ other than the cluster AZ, for single AZ clusters. This is then used to build a R53 Resolver to forward traffic to Qumulo DNS for Floating IP resolution.	`string`	`null`	no
region	AWS region for deployment	`string`	n/a	yes
s3_log_bucket_name	OPTIONAL: Bucket name for S3 logging	`string`	`null`	no
s3_log_bucket_prefix	OPTIONAL: Bucket prefix for S3 logging	`string`	`null`	no
soft_capacity_limit_tb	OPTIONAL: Soft capacity limit in TB (50 to 50000). Default is 500TB. Can be increased to add storage, but cannot be decreased. It's like a quota, unused capacity is not billed.	`number`	`500`	no
storage_class	HOT cluster default is INTELLIGENT_TIERING or STANDARD, COLD cluster default is GLACIER_IR or STANDARD_IA	`string`	`null`	no
subnet_ids	Subnet IDs for multi-AZ deployment (3+ subnets, one per AZ)	`list(string)`	n/a	yes
tags	OPTIONAL: Tags to apply to all AWS resources created for this cluster.	`map(string)`	`null`	no
vpc_id	VPC ID for the cluster	`string`	n/a	yes

Outputs

Name	Description
cluster_name	Name of the Qumulo cluster
cluster_soft_capacity_limit_tb	Total capacity the cluster may consume. Only used capacity is billed.
cluster_uuid	UUID of the Qumulo cluster

Name	Description
deployment_unique_name	Unique deployment identifier
endpoint_ips	Client-facing IPs. Floating IPs if configured, otherwise primary IPs.
endpoints	Connection endpoints for various protocols
primary_ips	Per-node primary IPs. Use these directly when no floating IPs are configured, or for per-node access.
provisioner_log	CloudWatch Log for the provisioner

About This Repository

For more information about this repository, contact Dack Busch and Gokul Kupparaj.